

Linear Systems on Graphs

Controllability of Structured Networks

Chapter 2

Foundations

Graph of a Linear System · Walks and the Controllability Matrix · Lin's Condition

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Chapter 2 Contents

2.1 State-Space Models and Their Graphs

2.2 Walks on the Graph

2.3 The Controllability Matrix

2.4 Structural Controllability on the Graph

The LTI State-Space Model: States, Inputs, Outputs

CONCEPT LTI State-Space Model

- ▶ Every system here: **linear, time-invariant (LTI), lumped** → a finite **state** vector $x(t) \in \mathbb{R}^n$

$$\dot{x} = Ax + Bu, \quad y = Cx + Du \quad (\text{almost always } D = 0)$$

- ▶ $x \in \mathbb{R}^n$ state, $u \in \mathbb{R}^m$ input, $y \in \mathbb{R}^q$ output. A, B, C, D **constant**
- ▶ x_{init} with u for $t \geq 0$ → the **entire future**. Nothing earlier needed

- ▶ Two descriptions → **internal** (the state, this course) and **external** (input → output, state eliminated → nothing to draw)

n states → n **state nodes** · m inputs → m **input nodes** · q outputs → q **output nodes**

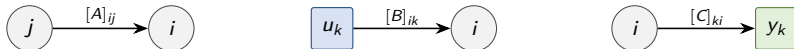
The Graph of a Linear System: One Node per State

CONCEPT Graph of a Linear System

$\dot{x} = Ax + Bu$, $y = Cx$ defines a **weighted directed graph** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^I \cup \mathcal{V}^O$ (one node per state, per input, per output) and

$$\begin{aligned}(j, i) \in \mathcal{E} &\iff [A]_{ij} \neq 0 \\ (u_k, i) \in \mathcal{E} &\iff [B]_{ik} \neq 0 \\ (i, y_k) \in \mathcal{E} &\iff [C]_{ki} \neq 0\end{aligned}$$

each edge carrying that element as its **weight**.



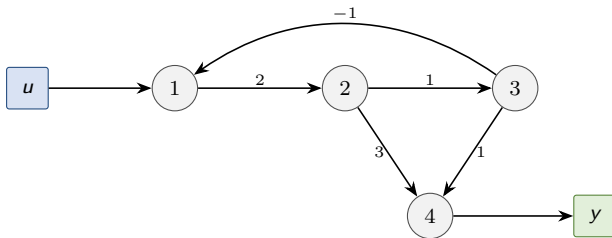
- **Direction:** $[A]_{ij} \neq 0 \rightarrow$ state j influences state $i \rightarrow$ edge $j \rightarrow i$
- **Column** index j influences, **row** index i is influenced
- $A =$ **weighted adjacency matrix**, transpose convention, fixed once
- Input edge $u \rightarrow i \rightarrow$ **the input enters node i**

Every linear system can be drawn. **What graph does a 4×4 matrix give?**

Matrix and Graph: A Four-Node Network

EXAMPLE Four-node network

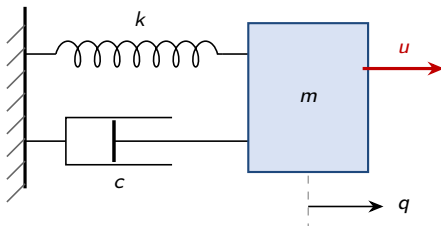
$$A = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}^T$$



16 elements vs. 5 edges → how the states interconnect is clearer in the graph

Mass, Spring and Damper: The Physical System

EXAMPLE Mass-spring-damper



q measured from equilibrium

$$m\ddot{q} = -kq - c\dot{q} + u$$

m mass, c damping, k stiffness, q position $\rightarrow$ the physical letters of this example, not the input and output dimensions m, q of $\dot{x} = Ax + Bu, y = Cx + Du$

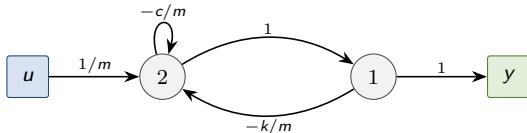
state $x = (q, \dot{q}) = (x_1, x_2)$, output = position $\rightarrow \dot{x}_1 = x_2, \quad \dot{x}_2 = -\frac{k}{m}x_1 - \frac{c}{m}x_2 + \frac{1}{m}u$

State = position and velocity. **What graph do these equations draw?**

From Equation to Graph: Mass-Spring-Damper

EXAMPLE Mass-spring-damper

$$A = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$



- ▶ $[A]_{12} = 1 \rightarrow$ edge $2 \rightarrow 1$
- ▶ $[A]_{21} = -k/m \rightarrow$ edge $1 \rightarrow 2$
- ▶ $[A]_{22} = -c/m \rightarrow$ edge $2 \rightarrow 2$, a **self-loop** (back to its own node)
- ▶ $[B]_{21} = 1/m \rightarrow$ edge $u \rightarrow 2$, input into node 2
- ▶ $[C]_{11} = 1 \rightarrow$ edge $1 \rightarrow y$

Equation and graph = one **model**. What does the self-loop mean?

The Self-Loop: What a Diagonal Element Means

CONCEPT Self-Loop

- ▶ $[A]_{ii} \neq 0 \rightarrow$ edge $i \rightarrow i \rightarrow$ state i feeds **itself** $\rightarrow$ the shortest **cycle**: one edge, start node = end node
- ▶ Weight = the rate at which state i drives **its own** change $\rightarrow$ negative = decay

EXAMPLE One self-loop, two domains

Mass-spring-damper

node 2 = velocity $\rightarrow [A]_{22} = -c/m$
 damper c dissipates, mass m stores.
 Spring off ($k = 0$): $\dot{x}_2 = -(c/m)x_2$,
 decay time m/c

RC circuit

one node = capacitor voltage $\rightarrow$
 $[A]_{11} = -1/(RC)$
 resistor R dissipates, capacitor C stores,
 decay time RC . Here C is capacitance, not
 the output matrix

Both examples $\rightarrow$ weight = $-(\text{dissipation})/(\text{storage}) \rightarrow$ the **storing** element sits underneath: m, C

Remove the damper and the self-loop is gone. **What is left of the motion?**

Structured Network: Erase the Weights

CONCEPT Pattern Graph

- ▶ Every element of A : a **fixed zero** (no edge) or a **free non-zero** parameter a_{ij} (edge, value unknown)
 - ▶ That matrix = **pattern matrix** $\Rightarrow$ **pattern graph**
 - ▶ One choice of the non-zero elements = a **realization** (Ch. 1)
-
- ▶ So far (A, B) carry both **edge weights** and **edge existence** $\rightarrow$ the **pattern matrices** keep **edge existence** only
 - ▶ Graph given $\rightarrow$ the **controllability matrix** built from it alone $\rightarrow$ the core of **Part II**

Part II $\rightarrow$ the graph read by **edge existence** only

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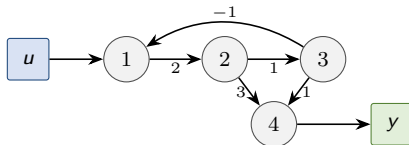
2.4 Structural Controllability on the Graph

Walk, Path, Cycle, Stem, Branch: Five Definitions

CONCEPT Five Definitions

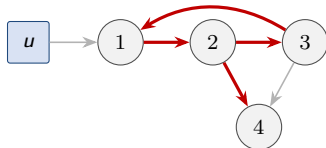
- ▶ **WALK** of length $k \rightarrow k$ consecutive directed edges, node repeats allowed.
Length 0 = empty sequence, a node to itself
- ▶ **PATH** $\rightarrow$ a walk, no node twice
- ▶ **CYCLE** $\rightarrow$ a walk with the **same start and end node**, no other node twice
- ▶ **STEM** $\rightarrow$ a path **starting from** an input node
- ▶ **BRANCH** $\rightarrow$ a path **ending at** an output node (dual of a stem)

EXAMPLE Four-node network

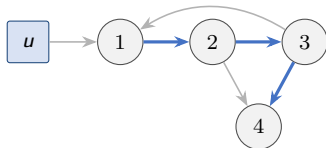


Among the 5, only the walk may repeat a node. **What do they look like on one graph?**

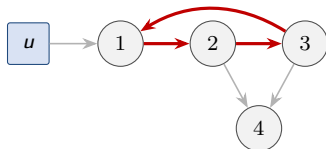
The Same Graph, Four Times: Walk, Path, Cycle, Stem



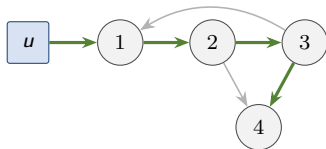
WALK of length 4
 $2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 4$, node 2 twice



PATH of length 3
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$, no node twice



CYCLE of length 3
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$, back to its start

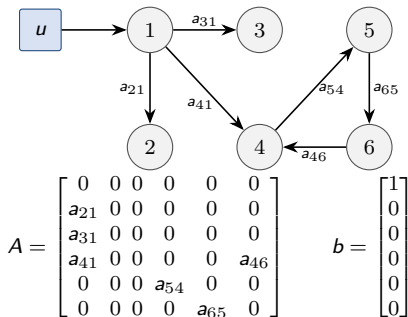


STEM out of the input node
 $u \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$, every state node

Only a cycle lets a walk repeat a node. **What matrix does a 6-node graph give?**

A Six-Node Network: One Input, One Cycle

EXAMPLE Network of the ECC 2024 paper, Fig. 1(b)

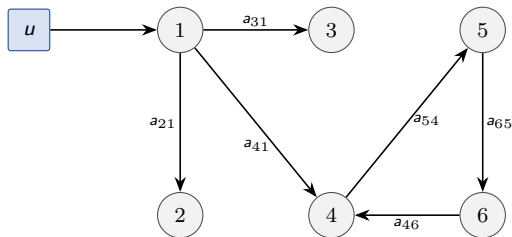


- A = the **adjacency matrix** of the graph: edge $j \rightarrow i$ carries a_{ij}

Edge existence fixed, edge weights free. **How is A read off the graph?**

Rows In, Columns Out: Reading the Six-Node Matrix

- ▶ Row i of $A \rightarrow$ the edges **entering** node i . Column $j \rightarrow$ the edges **leaving** node j
- ▶ Row 1 zero $\rightarrow$ no edge into node 1
- ▶ **Subscripts read backwards:** $[A]_{54} \rightarrow$ edge $4 \rightarrow 5$, **not** $5 \rightarrow 4$



Rows say what comes in, columns what goes out. **What do powers of A say?**

Powers of the Adjacency Matrix: $[A^k]_{lw}$

CONCEPT Weight Product, Sum of Weight Products

- ▶ **WEIGHT PRODUCT (WP)** of one walk $\rightarrow$ the product of the weights of its edges (six-node graph, walk $1 \rightarrow 4 \rightarrow 5 \rightarrow$ WP = $a_{54}a_{41}$)
- ▶ **SUM OF WEIGHT PRODUCTS (SWP)**, written $\text{SWP}_k(w \rightarrow l)$ $\rightarrow$ add the WPs of **every** k -step walk from w to l (one walk here $\rightarrow$ one WP. Two walks added $\rightarrow$ the diamond of §2.4)

▶ Walk Enumeration: The Key Identity of Part II

$$[A^k]_{lw} = \text{SWP}_k(w \rightarrow l) = \sum_{k\text{-step walks } w \rightarrow l} \text{WP}$$

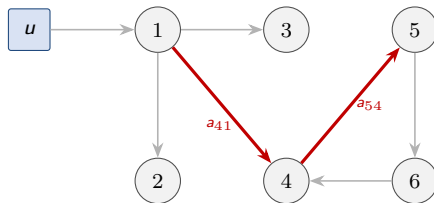
row l = walk **end** · column w = walk **start** · walk, so nodes may repeat · all weights 1 $\rightarrow$ a plain **count**

Every element of A^k is a sum over walks. **What is inside one element of A^2 ?**

A^2 Out of Node 1: One Walk, One Element

► The Structure of A^2

$$A^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{46}a_{65} & 0 \\ \textcolor{red}{a_{54}a_{41}} & 0 & 0 & 0 & 0 & a_{54}a_{46} \\ 0 & 0 & 0 & a_{65}a_{54} & 0 & 0 \end{bmatrix}$$

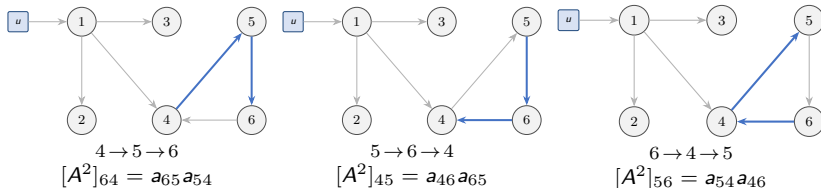


$1 \rightarrow 4 \rightarrow 5 \rightarrow \text{WP} = [A]_{54}[A]_{41} = \textcolor{red}{a_{54}a_{41}}$
 the only 2-step walk into node 5 $\rightarrow [A^2]_{51} = \text{SWP}_2(1 \rightarrow 5) = \textcolor{red}{a_{54}a_{41}}$

The product runs right to left, destination first. **Where else is A^2 non-zero?**

A^2 Inside the Cycle: Three More Elements

► Three Two-Step Walks Inside the Cycle

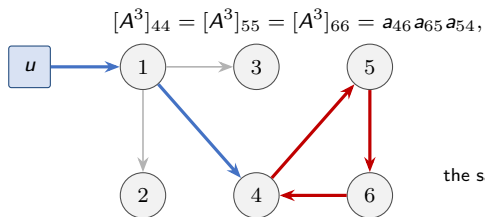


- 4 non-zero elements in A^2 , no others. Nodes 2 and 3 have no edge out $\rightarrow$ no walk of any length continues past them
- No diagonal element in $A^2 \rightarrow$ no closed walk of length 2. The one cycle here needs three steps

One walk out of node 1, 3 inside the cycle. **Where do the 3 land at $k = 3$?**

A^3 : The Cycle Lands on the Diagonal

► A^3 : Four Non-Zero Elements



One cycle, three starting nodes:

$$4 \rightarrow 5 \rightarrow 6 \rightarrow 4$$

$$5 \rightarrow 6 \rightarrow 4 \rightarrow 5$$

$$6 \rightarrow 4 \rightarrow 5 \rightarrow 6$$

the same three edges, the same product $\rightarrow$
three equal elements

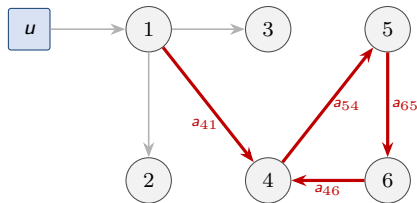
- $[A^k]_{ii} \neq 0 \iff$ a **closed walk** of length k through node i . The shortest such walk is a **cycle**
- Nodes 1, 2, 3: no diagonal element at any $k \rightarrow$ no cycle through them
- $[A^3]_{61} \rightarrow$ the walk $1 \rightarrow 4 \rightarrow 5 \rightarrow 6$, three steps out of node 1

Non-zero on the diagonal $\rightarrow$ node i sits on a cycle. **What does a fourth step add?**

A^4 : The Same Ends, Two Lengths

► Same Row, Same Column, Two Lengths

$$[A]_{41} = a_{41} \quad (1 \rightarrow 4, \text{ length } 1) \quad [A^4]_{41} = a_{46} a_{65} a_{54} a_{41} \quad (1 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 4, \text{ length } 4)$$



node 4 twice

$$\text{Cycle weight } P = a_{46} a_{65} a_{54} \rightarrow [A^4]_{41} = P a_{41}$$

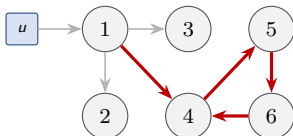
- A **walk**, not a path $\rightarrow$ node 4 at step 1 and again at step 4

Same start, same end, one extra lap $\rightarrow$ the factor P . **Which elements pick it up?**

One Lap Later: Every Element Times the Cycle Weight

► A^4 : Four Non-Zero Elements, One Lap Added to Each

$$[A^4]_{41} = P a_{41}, \quad [A^4]_{46} = P a_{46}, \quad [A^4]_{54} = P a_{54}, \quad [A^4]_{65} = P a_{65}$$



the four edges A^4 lands on

One lap $\rightarrow$ multiply by $P \rightarrow$ back on the length-1 positions whose end node has an edge out

- Non-zero again at $k = 7, 10, \dots$, one more lap each time: $[A^7]_{41} = P^2 a_{41}$
- The cycle runs again $\rightarrow A^k \neq 0$ for every $k \rightarrow$ the walk list never ends
- No cycle $\rightarrow$ every walk a path $\rightarrow$ at most n nodes $\rightarrow$ length $\leq n - 1 \rightarrow A^n = 0$

Controllability matrix = walks, one column per length. **How many columns are independent?**

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Rank: A Spanning Question

CONCEPT Rank

- ▶ **Column space** of $M \rightarrow$ all combinations of its columns
- ▶ Columns **linearly dependent** $\rightarrow$ one of them a combination of the others. Otherwise **linearly independent**
- ▶ $\text{rank } M = \text{number of independent columns} = \text{dimension of the column space} = \text{number of independent rows}$

EXAMPLE A worked 3×3

$$M = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \text{col } 3 = \text{col } 1 + \text{col } 2 \Rightarrow \text{columns dependent} \Rightarrow \text{rank } M = 2$$

- ▶ Three columns, one **plane**, not $\mathbb{R}^3 \rightarrow$ column 3 a combination of columns 1 and 2 $\rightarrow$ one redundant
- ▶ The dependency as a check $\rightarrow M \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}^\top = 0$

Next comes $\mathcal{C} = [b \quad Ab \cdots A^{n-1}b]$. **Do its columns span $\mathbb{R}^n$?**

The Controllability Matrix: One Column per Walk Length

$$\mathcal{C} = [b \quad Ab \quad A^2b \quad \dots \quad A^{n-1}b], \quad \text{controllable} \iff \text{rank } \mathcal{C} = n$$

Kalman rank condition (Ch. 1) · n = number of states

► The Mechanism

- **Single input:** $b = [1 \ 0 \ 0 \ 0 \ 0 \ 0]^T \rightarrow A^k b = \text{column 1 of } A^k$
- Element $(l, k+1)$ of $\mathcal{C} = \text{SWP}_k(1 \rightarrow l)$, the SWP of all k -step walks from **node 1** to node l
- The input edge carrying weight 1 $\rightarrow$ the count starting at node 1, not at u

	b	Ab	A^2b	A^3b	$\dots$	$A^{n-1}b$	
row = node l							
column = walk length k	0	1	2	3	$\dots$	$n-1$	

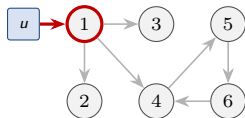
element $(l, k+1)$

Column $k+1$ = every k -step walk out of node 1. **What is in columns 1 and 2?**

Columns 1 and 2: b and Ab

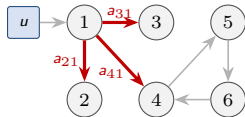
- **Column 1** = $b \rightarrow$ walk length 0 · node 1, where the input enters

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$



- **Column 2** = $Ab \rightarrow$ walk length 1 · $1 \rightarrow 2$, $1 \rightarrow 3$, $1 \rightarrow 4$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$

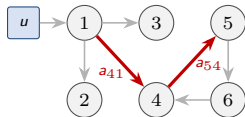


One step $\rightarrow$ nodes 2, 3 and 4. **Which nodes are still out of reach?**

Columns 3 and 4: A^2b and A^3b

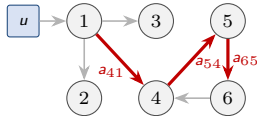
- **Column 3** = $A^2b \rightarrow$ walk length 2 · $1 \rightarrow 4 \rightarrow 5$

$$C = \begin{bmatrix} 1 & 0 & \mathbf{0} & 0 & 0 & 0 \\ 0 & a_{21} & \mathbf{0} & 0 & 0 & 0 \\ 0 & a_{31} & \mathbf{0} & 0 & 0 & 0 \\ 0 & a_{41} & \mathbf{0} & 0 & P a_{41} & 0 \\ 0 & 0 & \mathbf{a_{54} a_{41}} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & \mathbf{0} & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$



- **Column 4** = $A^3b \rightarrow$ walk length 3 · $1 \rightarrow 4 \rightarrow 5 \rightarrow 6$

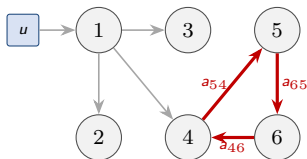
$$C = \begin{bmatrix} 1 & 0 & 0 & \mathbf{0} & 0 & 0 \\ 0 & a_{21} & 0 & \mathbf{0} & 0 & 0 \\ 0 & a_{31} & 0 & \mathbf{0} & 0 & 0 \\ 0 & a_{41} & 0 & \mathbf{0} & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & \mathbf{0} & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & \mathbf{a_{65} a_{54} a_{41}} & 0 & 0 \end{bmatrix}$$



Length 3 $\rightarrow$ the path $u \rightarrow 1 \rightarrow 4 \rightarrow 5 \rightarrow 6$ ends. **What do columns 5 and 6 add?**

Six Nodes, Six Columns: $\mathcal{C}$ in Full

$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$



$$P = a_{46} a_{65} a_{54}$$

$$P a_{41} = a_{46} a_{65} a_{54} a_{41}$$

P = the cycle $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$

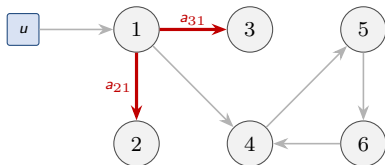
6 rows, 6 columns. **What is the rank, and which rows are dependent?**

Rank From the Graph: Two Nodes, One Ratio

► Where Does the Rank Stop, and Why?

- Row 2 = $[0 \ a_{21} \ 0 \ 0 \ 0 \ 0]$ and row 3 = $[0 \ a_{31} \ 0 \ 0 \ 0 \ 0]$, **proportional** for every realization $\Rightarrow \text{rank } C \leq 5 < 6$
- Column 2 the only column with a non-zero element in rows 2 and 3
 $\rightarrow$ one column carrying every walk to nodes 2 and 3

What if $a_{21} = a_{31}$ and desired $x_2 \neq x_3$?



one push out of node 1 $\rightarrow$ nodes 2 and 3 in the fixed ratio $a_{21} : a_{31}$

ratio $a_{21} : a_{31}$ fixed

The input never separates nodes 2 and 3. **How large can the rank be?**

Rank Exactly Five: The Determinant, Computed

► Drop Row 3 and Column 6: A 5×5 Is Left

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$

$P = a_{46} a_{65} a_{54}$, the cycle $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$

$$\det = a_{21} a_{41}^3 a_{54}^2 a_{65} a_{46} \neq 0 \implies \text{rank } C = 5$$

- One product of edge weights, every factor non-zero, nothing to cancel against $\rightarrow$ rank $C = 5$ for **every** realization (no exceptional choice of weights $\rightarrow$ §2.4)
- No weight chosen, no number measured, only which elements are non-zero $\rightarrow$ the rank fixed by **edge existence**, not by **edge weight**

Rank 5 from a determinant. **Can the graph alone give the same 5?**

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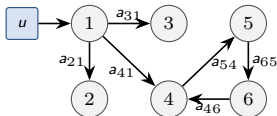
2.4 Structural Controllability on the Graph

Structural Controllability: One Pattern, Many Realizations

CONCEPT Structural Controllability (SC)

- ▶ **PATTERN** $(A, B) \rightarrow$ which elements are free, which are **fixed zero**. No number chosen yet
- ▶ **REALIZATION** $F \rightarrow$ one set of numbers for the free elements $\rightarrow (A_F, B_F)$
- ▶ **FAMILY** $\rightarrow$ all the realizations of one pattern
- ▶ **Structural Controllability (SC)** $\rightarrow$ controllable for **almost all** realizations of the pattern (the exceptions = a measure-zero set of weights)

EXAMPLE Network of the ECC 2024 paper, Fig. 1(b)



6 free weights, one pattern $\rightarrow$ one family
 rank $\mathcal{C}_F = 5 < 6$ at **every** realization
 no realization controllable $\rightarrow$ **not SC**

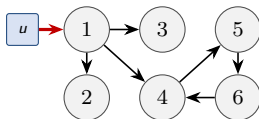
The input reaches every node here. **What if one node were out of reach?**

Reachability: Three Definitions

CONCEPT Three Definitions

- ▶ **REACHABLE** state node $\rightarrow$ a node reached by some stem from an input
- ▶ **INPUT-CONNECTED** $\rightarrow$ every state node reachable from an input
- ▶ **REACHABLE SET** $\rightarrow$ the set of reachable state nodes

EXAMPLE ECC 2024 Fig.1(b) · a 3-node pattern



every state node reachable

$\rightarrow$ **input-connected**, reachable set = all 6



no edge out of node 1

$\rightarrow$ nodes 2, 3 **unreachable**

reachable set = $\{1\}$ $\rightarrow$ **not input-connected**

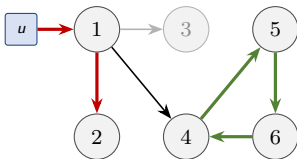
Nodes 2 and 3 are drawn, yet unreachable. **How is the rank read off the graph?**

Covering the Graph: Stems and Cycles, No Node Shared

CONCEPT Cover

- ▶ **COVER** → a set of stems and cycles sharing **no node** (a *disjoint* set)
- ▶ **COVERED** state nodes → the **state** nodes those stems and cycles run through (*u* not counted)
- ▶ **MAXIMUM COVER** → the maximum number of state nodes any such set covers

EXAMPLE Network of the ECC 2024 paper, Fig. 1(b)



STEM $u \rightarrow 1 \rightarrow 2 \rightarrow$ nodes 1, 2

CYCLE $4 \rightarrow 5 \rightarrow 6 \rightarrow 4 \rightarrow$ nodes 4, 5, 6

Maximum cover 5, and rank 5. **Is that a coincidence or a rule?**

Lin's Condition: Reach Every Node, Cover Every Node

THEOREM Lin

A graph is **SC** if and only if

- (C1) **input-connected**
- (C2) a **disjoint set of stems and cycles** covers all state nodes

THEOREM Hosoe

$\text{rank } \mathcal{C} =$ the **maximum** number of **reachable** state nodes covered by (C2)

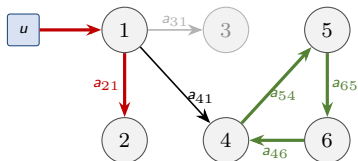
► Where Each Count Comes From

- **Node not reachable** → no walk from the input ends there → that row of $\mathcal{C}$ zero at every realization
- **Node not covered** → more state nodes than the nodes feeding them → those rows of $\mathcal{C}$ have rank below their count at every realization

Two counts, no weight chosen. **Does the cover match the rank on every graph?**

The Six-Node Network: Cover 5, Rank 5

EXAMPLE Network of the ECC 2024 paper, Fig. 1(b)



STEM $u \rightarrow 1 \rightarrow 2$, **CYCLE** $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$
cover 5 of 6, node 3 left out
 reachable set = all 6 $\rightarrow$ maximum cover
 on it = 5 $\rightarrow$ $\text{rank } C = 5$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & a_{54} a_{41} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & a_{65} a_{54} a_{41} & 0 & 0 \end{bmatrix}$$

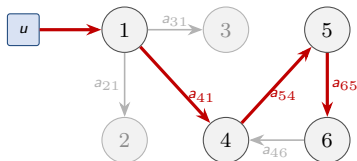
$P = a_{46} a_{65} a_{54}$, the cycle $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$

rows 2 and 3 proportional at every realization $\rightarrow \text{rank } C \leq 5 < 6 \rightarrow$ **not SC**

Cover 5 of 6, rank 5. **Which rows does the stem alone account for?**

One Stem, Four Rows: Rank 4 From the Stem Alone

EXAMPLE The same network · a longer stem



STEM $u \rightarrow 1 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow$ nodes 1, 4, 5, 6

one node reached at each walk length

0, 1, 2, 3

$\rightarrow$ rows 1, 4, 5, 6 of C : a **staircase**

$\rightarrow$ 4 **independent** rows $\rightarrow \text{rank } C \geq 4$

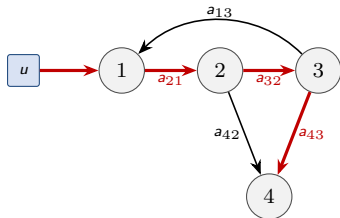
$$C = \begin{bmatrix} \mathbf{1} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 & 0 \\ 0 & \mathbf{a_{41}} & 0 & 0 & P a_{41} & 0 \\ 0 & 0 & \mathbf{a_{54} a_{41}} & 0 & 0 & P a_{54} a_{41} \\ 0 & 0 & 0 & \mathbf{a_{65} a_{54} a_{41}} & 0 & 0 \end{bmatrix}$$

the cycle $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$ shares the stem's nodes $\rightarrow$ nothing to add here
 the stem $u \rightarrow 1 \rightarrow 2$ with that cycle $\rightarrow$ 5 nodes, 5 rows $\rightarrow \text{rank } C = 5$ (previous slide)

One stem, 4 independent rows. **What if one stem covers every node?**

The Four-Node Network: One Stem, 4 of 4

EXAMPLE Four-node network of §2.1 · Book, fig:pet-pattern



STEM $u \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow$ all 4 nodes
 edges $3 \rightarrow 1$ and $2 \rightarrow 4$ off the stem
cover 4 of 4
 reachable set = all 4 $\rightarrow$ rank $\mathcal{C} = 4$

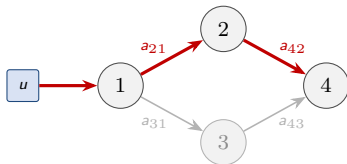
$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 & a_{13}a_{21}a_{32} \\ 0 & a_{21} & 0 & 0 \\ 0 & 0 & a_{21}a_{32} & 0 \\ 0 & 0 & a_{21}a_{42} & a_{21}a_{32}a_{43} \end{bmatrix}$$

every non-zero element a single product, and $\det \mathcal{C} = a_{21}^3 a_{32}^2 a_{43}$, one product again
 $\rightarrow$ rank $\mathcal{C} = 4$, **SC**

Cover 4 of 4, 2 edges to spare. **What if two nodes feed the same node?**

The Diamond: Cover 3 of 4, Rank 3

EXAMPLE Book, ex:diamond



STEM $u \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow$ nodes 1, 2, 4
 no cycle $\rightarrow$ **cover 3 of 4**, node 3 left out
 every state node reachable $\rightarrow$ **input-connected**
 reachable set = all 4 $\rightarrow$ rank $C = 3$

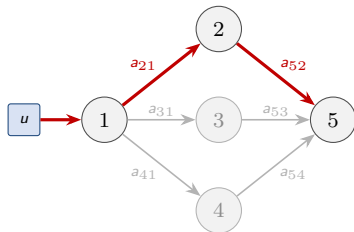
$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 \\ 0 & a_{31} & 0 & 0 \\ 0 & 0 & a_{42}a_{21} + a_{43}a_{31} & 0 \end{bmatrix}$$

column 4 zero at every realization $\rightarrow$ rank $C = 3 < 4 \rightarrow$ **not SC**

Cover 3 of 4, rank 3, node 3 left out. **What if 3 nodes sit in the middle?**

The Fan: Cover 3 of 5, Rank 3

EXAMPLE Book, ex:fan



STEM $u \rightarrow 1 \rightarrow 2 \rightarrow 5 \rightarrow$ nodes 1, 2, 5
 no cycle $\rightarrow$ **cover 3 of 5**, nodes 3, 4 left out
 every state node reachable $\rightarrow$ **input-connected**
 reachable set = all 5 $\rightarrow$ rank $\mathcal{C} = 3$

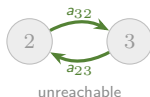
$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 & 0 \\ 0 & a_{31} & 0 & 0 & 0 \\ 0 & a_{41} & 0 & 0 & 0 \\ 0 & 0 & a_{52}a_{21} + a_{53}a_{31} + a_{54}a_{41} & 0 & 0 \end{bmatrix}$$

columns 4 and 5 zero at every realization $\rightarrow$ rank $\mathcal{C} = 3 < 5 \rightarrow$ **not SC**

Cover 3 of 5, rank 3, the count matching again. **Can the cover ever exceed the rank?**

Covered but Unreachable: Cover 3, Rank 1

EXAMPLE Book, ex:covered-inaccessible · covered, yet unreachable



STEM $u \rightarrow 1 \rightarrow \text{node 1}$

CYCLE $2 \rightarrow 3 \rightarrow 2 \rightarrow \text{nodes 2, 3, no node shared}$
 $\rightarrow$ **cover 3 of 3**

no edge out of node 1 $\rightarrow$ nodes 2, 3 **unreachable, not input-connected**

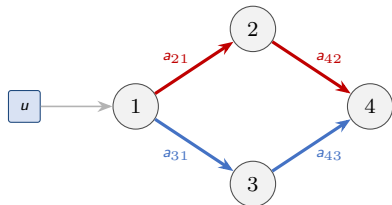
$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

rows 2 and 3 zero at every realization $\rightarrow$ maximum cover on the reachable set = 1 $\rightarrow$
rank $C = 1 < 3$, not SC

Cover on the reachable set = rank. Is every element of C one product?

One Element, Two Walks: The Sum Inside $\mathcal{C}$

EXAMPLE Book, ex:diamond



walk $1 \rightarrow 2 \rightarrow 4 \rightarrow$ product $a_{42}a_{21}$
 walk $1 \rightarrow 3 \rightarrow 4 \rightarrow$ product $a_{43}a_{31}$
 both of length 2 $\rightarrow$ both landing in
 column 3 of $\mathcal{C}$, the A^2b column

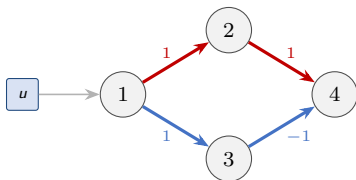
$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a_{21} & 0 & 0 \\ 0 & a_{31} & 0 & 0 \\ 0 & 0 & a_{42}a_{21} + a_{43}a_{31} & 0 \end{bmatrix} \quad \text{column 4} = A^3b = 0: \text{ no edge out of node 4}$$

$$[\mathcal{C}]_{43} = a_{42}a_{21} + a_{43}a_{31}$$

One element, two walks, one sum. **Which weights make the sum 0?**

One Sum, One Cancellation: The Rank Drops

EXAMPLE The diamond again · one choice of weights



$$a_{21} = a_{31} = a_{42} = 1, \quad a_{43} = -1$$

$$[C]_{43} = a_{42}a_{21} + a_{43}a_{31} = 1 - 1 = 0$$

$$C_F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{rank } C_F = 2 \quad \text{cover 3, and rank 3 at almost every other choice}$$

The fan cancels the same way: $a_{52}a_{21} + a_{53}a_{31} + a_{54}a_{41} = 1 + 1 - 2 = 0$

One choice of weights $\rightarrow$ rank below the cover $\rightarrow$ **how rare is such a choice?**

Almost All, Not All: The Door to SSC

CONCEPT Two Words for Controllable

- ▶ **Structural Controllability (SC)** → controllable for **almost all** realizations.
The cancelling weights: a measure-zero set, left out
- ▶ **Strong Structural Controllability (SSC)** → controllable for **all** realizations,
the cancelling weights included

Ch. 5

▶ What the Cover Count Promises

- ▶ Cover = rank → a statement about **almost all** weights (Lin, Hosoe)
- ▶ A sum inside $\mathcal{C}$ → some weights make it 0 → the rank drops **at those weights**
- ▶ Six-node network: every element one product → nothing to cancel → rank 5
at **every** realization

Sum = 0 → rank drops → **counting those weights in = SSC**

Summary and Next Week: Ch.3 Controllability

► Chapter 2 in One Breath

state model → its **graph** → walks and powers → controllability matrix → **cover = rank**

- Linear system = weighted directed graph → elements of A^k and $\mathcal{C}$ = **walks** on it
- rank $\mathcal{C}$ read off the graph → **input-connected** + the **maximum stem-and-cycle cover** (Lin)
- Almost all realizations, **not all** → the cancelling ones open the door to **SSC** (Ch. 5)

Six-node: cover 5 = rank 5, not SC · four-node: 4 = 4, SC · diamond: a sum inside $\mathcal{C}$, rank 2 at one choice of weights

[Next week] Ch.3, the Kalman rank condition → **can every state be steered?**